20.1, 20.2 Stokes' Theorem

If $\vec{F} = \langle F_1, F_2, F_3 \rangle$ is a vector field in $\mathbb{R}^3$, then its (vector) curl is given by

$$\text{curl} = \nabla \times \vec{F}.$$

Imagine that you put a paddle-wheel with a handle at a point in the field. Imagine that the field can rotate the paddle-wheel. Now find the angle of the handle which makes the paddle-wheel rotate the fastest. Then $\nabla \times \vec{F}$ points in that direction, and $\|\nabla \times \vec{F}\|$ is how "fast" the paddle-wheel is rotating (i.e., the curl!).

→ Do problem 1.

Green's Theorem in three dimensions is:

<u>Stokes' Theorem</u> If S is a smooth oriented surface with piecewise smooth oriented boundary C, and if $\vec{F}$ is a smooth vector field in $\mathbb{R}^3$, then

$$\oint_C \vec{F} \cdot \partial \vec{r} = \int_S (\nabla \times \vec{F}) \cdot \partial \vec{A}.$$

The orientation on C is determined by the orientation on S by the right-hand rule.

→ Do rest of problems.